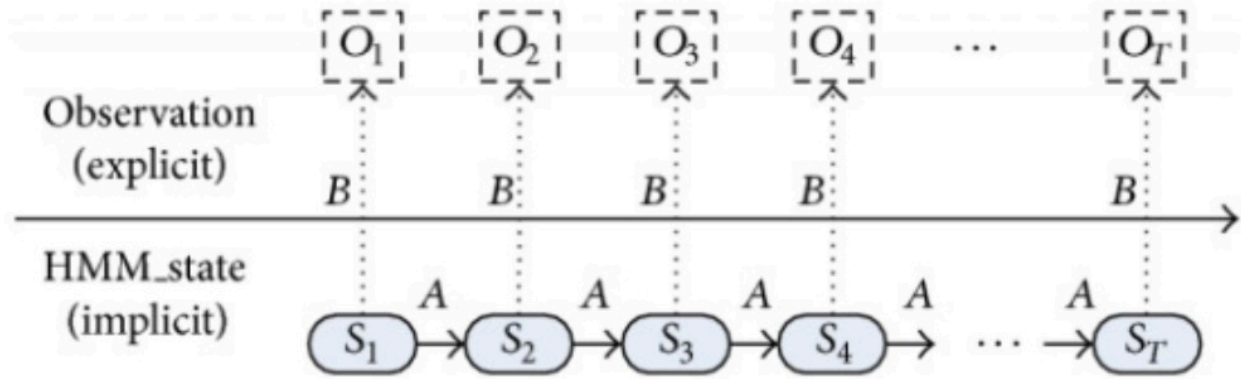


Hidden Markov Models (HMM)

What is an HMM?



An HMM has two layers of reality:

- **Observations (O):** Things you can actually see/hear (e.g., the speech signal)
- **Hidden States (Q):** The underlying reality you can't see (e.g., which phoneme is being spoken)

The model is defined by three parameters: $\lambda = (A, B, \pi)$

- **A = Transition matrix** $\rightarrow$ probability of going from state i to state j
- **B = Emission matrix** $\rightarrow$ probability of observing symbol o when in state i
- **π = Initial state distribution** $\rightarrow$ probability of starting in each state

HMM Fundamental Problems

1. Evaluation $\rightarrow$ compute $P(O|\lambda)$

Given:

- Observations
- Model: $\lambda = (A, B, \pi)$

2. Learning $\rightarrow$ compute model parameters

Given: Observation sequence

3. Decoding $\rightarrow$ find best hidden states

Direct Computation of Evaluation Problem

1) Observation likelihood given state sequence and model

$$P(O | Q, \lambda) = b_1(o_1) \cdot b_2(o_2) \cdot b_3(o_3) \dots b_T(o_T)$$

2) State sequence probability given model

$$P(Q | \lambda) = \pi_1 \cdot a_{12} \cdot a_{23} \cdot a_{34} \dots a_{(T-1)T}$$

3) Joint probability

$$P(O, Q | \lambda) = \pi_{q_1} \cdot b_{q_1}(o_1) \cdot \prod a_{q(t-1)q_t} \cdot b_{q_t}(o_t)$$

Simple way:

$$\pi(q_1) \times B(q_1, o_1) \times A(q_1 \rightarrow q_2) \times B(q_2, o_2) \times \dots \times A(q_{(T-1)} \rightarrow q_T) \times B(q_T, o_T)$$

Part 2 — The Forward Algorithm

The Forward Algorithm is about the evaluation problem in Hidden Markov Models (HMM).

We are given a model:

$$\lambda = (A, B, \pi)$$

And we want to compute:

$$P(O | \lambda)$$

Meaning: *What is the probability of observing a sequence O given the model?*

Key Idea

$\alpha_t(i)$ is the probability of seeing the observations up to time t and being in state i at time t.

$$\alpha_t(i) = P(o_1, o_2, \dots, o_t, \text{state } i \text{ at time } t | \lambda)$$

Steps of Forward Algorithm

(1) Initialization (t = 1)

$$\alpha_1(i) = \pi_i \cdot b_i(o_1)$$

(2) Induction (t → t+1)

$$\alpha_{t+1}(j) = \left(\sum \alpha_t(i) \cdot a_{ij} \right) \cdot b_j(o_{t+1})$$

- Come from all previous states
- Multiply by transition probability
- Multiply by emission of new observation

(3) Termination

$$P(O | \lambda) = \sum \alpha_T(i)$$

Solve Your Numerical Example

Initial Probabilities:

$$\pi = [0.5, 0.5]$$

Transition Matrix A:

$$A = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

Emission Matrix B:

$$B = \begin{bmatrix} 0.5 & 0.4 \\ 0.1 & 0.3 \end{bmatrix}$$

HMM Forward Algorithm – Full Solution

Step 1: Initialization

$$\alpha_1(1) = \pi_1 \cdot b_1(o_1) = 0.5 \cdot 0.5 = 0.25$$

$$\alpha_1(2) = \pi_2 \cdot b_2(o_1) = 0.5 \cdot 0.1 = 0.05$$

Step 2: Induction (t = 2)

Compute $\alpha_2(1)$

$$\begin{aligned}\alpha_2(1) &= [(0.25 \cdot 0.6) + (0.05 \cdot 0.3)] \cdot 0.4 \\ &= (0.15 + 0.015) \cdot 0.4 = 0.165 \cdot 0.4 = 0.066\end{aligned}$$

Compute $\alpha_2(2)$

$$\begin{aligned}\alpha_2(2) &= [(0.25 \cdot 0.4) + (0.05 \cdot 0.7)] \cdot 0.3 \\ &= (0.1 + 0.035) \cdot 0.3 = 0.135 \cdot 0.3 = 0.0405\end{aligned}$$

Step 3: Termination

$$\begin{aligned}P(O \mid \lambda) &= \alpha_2(1) + \alpha_2(2) \\ &= 0.066 + 0.0405 = 0.1065\end{aligned}$$

Final Answer: $P(O \mid \lambda) = 0.1065$
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Part 3 — The Decoding Problem

What is the Decoding Problem?

Given:

- Observations
- Model: $\lambda = (A, B, \pi)$

Find:

- Best hidden states sequence: $Q = (q_1, q_2, \dots, q_T)$

Viterbi Algorithm

Define the highest probability of a state sequence ending in state i at time t :

$$\delta_t(i) = \max P(q_1, q_2, \dots, q_{t-1}, q_t = i, o_1, o_2, \dots, o_t \mid \lambda)$$

Viterbi Algorithm Steps

(1) Initialization

The highest probability along a single path, at time 1, which accounts for the first observation o_1 and ends in state i .

$$\delta_1(i) = \pi_i \cdot b_i(o_1), \quad 1 \leq i \leq N$$

(2) Induction

$$\delta_{t+1}(j) = \max [\delta_t(i) \cdot a_{ij}] \cdot b_j(o_{t+1}), \quad 1 \leq j \leq N$$

(3) Termination

$$P^* = \max \delta_T(i)$$
$$q^{*T} = \arg \max \delta_T(i)$$

A Solved Question – Viterbi

Given:

- $N = 2$ states: State 1 (S1), State 2 (S2)
- $T = 3$ observations: $O = (A, B, A)$
- $\pi = [0.6, 0.4]$ (initial probabilities)
- Transition matrix A:
 $a_{11} = 0.7, a_{12} = 0.3$
 $a_{21} = 0.4, a_{22} = 0.6$
- Emission matrix B (symbol A, symbol B):
 $b_1(A) = 0.5, b_1(B) = 0.5$
 $b_2(A) = 0.4, b_2(B) = 0.6$

(1) Initialization — $t = 1$, observation = A

Rule: $\delta_1(i) = \pi_i \times b_i(O_1), \psi_1(i) = 0$

Apply:

$$\delta_1(1) = \pi_1 \times b_1(A) = 0.6 \times 0.5 = 0.30 \rightarrow \psi_1(1) = 0$$

$$\delta_1(2) = \pi_2 \times b_2(A) = 0.4 \times 0.4 = 0.16 \rightarrow \psi_1(2) = 0$$

(2) Induction — $t = 2$, observation = B

For state $j = 1$

Rule: $\max_i [\delta_{t-1}(i) \times a_{ij}] \rightarrow$ Choose max transition:

$$\delta_1(1) \times a_{11} = 0.30 \times 0.7 = 0.21 \leftarrow \max$$

$$\delta_1(2) \times a_{21} = 0.16 \times 0.4 = 0.064$$

Rule: $\delta_t(j) = \max_i [\delta_{t-1}(i) \cdot a_{ij}] \times b_j(O)$

$$\delta_2(1) = 0.21 \times b_1(B) = 0.21 \times 0.5 = 0.105$$

Backpointer: $\psi_2(1) = 1$

For state $j = 2$

Rule: $\max_i [\delta_{t-1}(i) \times a_{ij}] \rightarrow$ Choose max transition:

$$\delta_1(1) \times a_{12} = 0.30 \times 0.3 = 0.09$$

$$\delta_1(2) \times a_{22} = 0.16 \times 0.6 = 0.096 \leftarrow \max$$

Rule: $\delta_t(j) = \max_i [\delta_{t-1}(i) \cdot a_{ij}] \times b_j(O)$

$$\delta_2(2) = 0.096 \times b_2(B) = 0.096 \times 0.6 = 0.0576$$

Backpointer: $\psi_2(2) = 2$

Induction — $t = 3$, observation = A

For state $j = 1$

$$0.105 \times 0.7 = 0.0735 \leftarrow \max$$

$$0.0576 \times 0.4 = 0.023$$

Rule: $\delta_t(j) = \max_i [\delta_{t-1}(i) \cdot a_{ij}] \times b_j(O)$

$$\delta_3(1) = 0.0735 \times b_1(A) = 0.0735 \times 0.5 = 0.03675$$

Backpointer: $\psi_3(1) = 1$

For state j = 2

$$0.105 \times 0.3 = 0.0315$$

$$0.0576 \times 0.6 = 0.03456 \leftarrow \max$$

Rule: $\delta_t(j) = \max_i [\delta_{t-1}(i) \cdot a_{ij}] \times b_j(O)$

$$\delta_3(2) = 0.03456 \times b_2(A) = 0.03456 \times 0.4 = 0.01382$$

Backpointer: $\psi_3(2) = 2$

(3) Termination

Rule: $P^* = \max_i \delta_T(i)$, $q^*T = \operatorname{argmax}_i \delta_T(i)$

$$P^* = \max(0.03675, 0.01382) = 0.03675$$

$$q^*_3 = 1$$

Step 4 — Backtracking

Rule: $q^*t = \psi_{t+1}(q^*_{t+1})$

$$q^*_3 = 1$$

$$q^*_2 = \psi_3(1) = 1$$

$$q^*_1 = \psi_2(1) = 1$$

Best Path: S1 → S1 → S1 | P* = 0.03675